



Malmalee Creations

SRS Documentation

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Chapter : 1 Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define the functional and non-functional requirements of the Handmade Items & Clothing E-Commerce Website. The system enables customers to browse, purchase, and track handmade products and clothing online, while allowing administrators to efficiently manage products, categories, customers, orders, and inventory.

1.2 Scope

The Handmade Items & Clothing E-Commerce Website is a web-based application designed to provide a secure and user-friendly online shopping experience. Customers can create accounts, browse products, search items, manage shopping carts, make payments, and track their orders. Administrators can manage products, categories, customers, orders, inventory, and generate reports through an administrative dashboard.

Main Features

- Customer registration and login
- Product browsing and searching
- Shopping cart management
- Secure checkout and online payment
- Order tracking
- Product reviews and ratings
- Admin dashboard for system management

Chapter : 2 Overall Description

2.1 Product Perspective

The Handmade Items & Clothing E-Commerce Website is a standalone web application developed using the MERN Stack. It provides a responsive interface for customers to shop online and a secure admin panel for managing the business.

Frontend

- React.js – Builds the interactive user interface.
- Tailwind CSS – Creates a responsive and modern design.
- Axios – Handles communication between the frontend and backend APIs.

Backend

- Node.js – Provides the server-side runtime environment.

- Express.js – Manages API routes and business logic.

Database

- MongoDB Atlas – Stores user accounts, products, orders, and other application data in the cloud.

Chapter : 3 Functional Requirements

3.1 Customer Registration and Authentication

- The system shall allow a guest to register a new account using name, email address, and password.
- The system shall validate that the email address is unique before creating a new account.
- The system shall hash and securely store customer passwords; passwords shall never be stored in plain text.
- The system shall allow a registered customer to log in using their email and password.
- The system shall issue a JWT (JSON Web Token) upon successful login to authenticate subsequent requests.
- The system shall allow a customer to log out, invalidating their active session on the client.
- The system shall allow a customer to reset a forgotten password via a verification link sent to their registered email.
- The system shall allow a customer to view and update their profile information, including saved delivery address and contact number.

3.2 Product Browsing, Search and Filtering

- The system shall display all active product categories on the website home page and navigation menu.
- The system shall allow a user to view all products belonging to a selected category.
- The system shall allow a user to search for products by name or keyword.
- The system shall allow a user to filter product listings by price range and availability (in stock / out of stock).
- The system shall allow a user to sort product listings by price (low to high, high to low) and by newest arrivals.

- The system shall display, for each product, a thumbnail image, name, price, and short description in listing views.

3.3 Product Detail Page

- The system shall display a dedicated detail page for each product, including full description, price, category, and stock availability.
- The system shall display a gallery of one or more product images on the product detail page.
- The system shall display an embedded short video clip on the product detail page when the admin has uploaded one for that product.
- The system shall allow the customer to select a quantity and add the product to the shopping cart from the detail page.
- The system shall display related or suggested products from the same category on the product detail page.

3.4 Shopping Cart

- The system shall allow a user to add a product to the shopping cart, specifying quantity.
- The system shall allow a user to update the quantity of an item in the cart or remove an item entirely.
- The system shall automatically recalculate the cart subtotal whenever items or quantities are changed.
- The system shall persist cart contents for a logged-in customer across sessions.
- The system shall display a cart summary, including item subtotal, delivery charge (if applicable), and total amount payable.

3.5 Checkout, Payment Gateway and Cash on Delivery

- The system shall require the customer to provide or confirm a delivery address, recipient contact number, and preferred delivery date before checkout is completed.
- The system shall allow the customer to choose a payment method at checkout: online payment (card / digital wallet) or Cash on Delivery (COD).
- When online payment is selected, the system shall redirect the customer to a secure, integrated third-party payment gateway to complete the transaction.

- The system shall create the order only after receiving a successful payment confirmation from the payment gateway, for online payments.
- When Cash on Delivery is selected, the system shall create the order immediately with a payment status of “Pending – COD” and shall not require online payment.
- The system shall generate a unique order number and display an order confirmation page/receipt after successful order placement.
- The system shall send an order confirmation notification (email) to the customer after the order is placed.
- The system shall handle and display a clear error message if a payment attempt fails, and shall allow the customer to retry payment or switch to COD.

3.6 Order History and Tracking (Customer)

- The system shall allow a registered customer to view a list of their past and current orders.
- The system shall allow a customer to view the detailed status of a specific order (e.g., Pending, Confirmed, Processing, Out for Delivery, Delivered, Cancelled).
- The system shall allow a customer to cancel an order only while its status remains “Pending” or “Confirmed.”

3.7 Admin Authentication and Access Control

- The system shall provide a separate, secured login for administrators, distinct from the customer login.
- The system shall restrict access to all admin-panel routes and API endpoints to authenticated administrator accounts only.
- The system shall automatically log out an admin session after a defined period of inactivity.

3.8 Admin – Category Management (CRUD)

- The system shall allow the admin to create a new product category by providing a category name, description, and image.
- The system shall allow the admin to view a list of all existing categories.
- The system shall allow the admin to edit the name, description, or image of an existing category.
- The system shall allow the admin to delete a category, provided no active products are currently assigned to it, or shall require reassignment of its products first.

- The system shall allow the admin to activate or deactivate a category, hiding deactivated categories from the public website.

3.9 Admin – Product Management (CRUD) and Media Upload

- The system shall allow the admin to create a new product with name, description, price, stock quantity, assigned category, and optional attributes such as color and material (e.g., ribbon, lace, fabric).
- The system shall allow the admin to upload one or more images for a product.
- The system shall allow the admin to upload a short video clip (e.g., MP4, up to a defined maximum file size such as 50 MB) for a product, to be displayed on the product detail page.
- The system shall allow the admin to view a searchable, paginated list of all products.
- The system shall allow the admin to edit any attribute of an existing product, including replacing or removing its images and video clip.
- The system shall allow the admin to delete a product, subject to confirmation, and shall prevent the product from appearing on the public site once deleted or deactivated.
- The system shall allow the admin to mark a product as “featured” so that it appears in a highlighted section on the website home page.
- The system shall automatically update product stock quantity when an order containing that product is placed.

3.10 Admin – Order Management

- The system shall allow the admin to view a list of all orders, filterable by status, payment method, and date range.
- The system shall allow the admin to view full order details, including customer information, items ordered, delivery address, and payment status.
- The system shall allow the admin to update the status of an order (e.g., Confirmed, Processing, Out for Delivery, Delivered, Cancelled).
- The system shall allow the admin to mark a Cash on Delivery order as “Paid” once payment has been collected upon delivery.

3.11 Admin Dashboard – Overview

- The system shall display, on the admin dashboard home page, summary statistics including total orders, total revenue, pending orders, and low-stock products.

- The system shall display a list of recently placed orders on the admin dashboard for quick access.

Chapter : 4 External Interface Requirements

4.1 User Interfaces

The public website shall provide an intuitive, visually appealing, mobile-responsive interface reflecting a handcrafted, boutique-style aesthetic, with clear navigation between the home page, category listings, product detail pages, cart, and checkout flow. The admin panel shall provide a clean, functional dashboard interface optimized for desktop use, organized into a sidebar navigation (Dashboard, Categories, Products, Orders) and a main content area for each management screen.

4.2 Hardware Interfaces

The system does not require any special-purpose hardware. It shall operate on standard consumer devices (desktop computers, laptops, tablets, and smartphones) with an internet connection and a supported web browser.

4.3 Software Interfaces

- Payment Gateway API – for processing online card/wallet payments and receiving payment confirmation callbacks.
- Email/SMTP Service – for sending order confirmation and password reset emails.
- Cloud/Local File Storage – for storing uploaded product images and video clips.
- Relational Database (MongoDB) – for persistent storage of all application data, accessed via the backend ORM.

4.4 Communication Interfaces

All communication between the frontend applications (customer website and admin panel) and the backend API shall occur over HTTPS using RESTful JSON APIs. Communication with the payment gateway shall use the gateway provider's secure API/redirect flow in accordance with its documentation.

Chapter : 5 Non-Functional Requirements

5.1 Performance

- The system shall load the product listing page within 3 seconds under normal network conditions.
- The system shall support at least 100 concurrent users browsing the website without significant degradation in response time.

5.2 Security

- The system shall enforce HTTPS for all client-server communication.
- The system shall never store raw payment card details; all card handling shall occur through the payment gateway's secure interface.
- The system shall hash all stored passwords using a strong, salted hashing algorithm (e.g., bcrypt).
- The system shall validate and sanitize all user-supplied input on both the client and server to mitigate injection attacks.

5.3 Usability

- The public website shall be responsive and usable on screen widths from 360px (mobile) to 1920px (desktop).
- The system shall display clear, human-readable error and validation messages for all forms.

5.4 Reliability and Availability

- The system shall target at least 99% uptime, excluding scheduled maintenance.
- The system shall retain uploaded product images and video clips reliably, with no data loss during normal operation.

5.5 Maintainability and Scalability

- The backend shall be structured using a layered/MVC architecture to support future maintenance and feature additions.
- The system's database schema shall support the addition of new product categories and attributes without requiring major architectural changes.

